Homework 4: Financial Data Analysis

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Due on April 26, 2026 (11:59pm)

Submit an electronic version of your answer (report and codes files) to Blackboard (late submission will not be accepted).

1. Fit AR(0)-ARCH(p) models with various orders p to the daily Bitcoin returns. Comment on the fit of the models. How many lags are needed to achieve a good fit? For fitted AR(0)-ARCH(1) model,
   (a) Calculate the unconditional mean, variance and kurtosis of the daily Bitcoin returns.
   (b) Plot the estimated conditional volatility of the daily Bitcoin returns.
   (c) Is it necessary to replace the AR(0) for the conditional mean with some alternative ARMA model?
2. Fit a AR(0)-GARCH(1,1) model to the daily Bitcoin returns. Compare the fit of the model with the AR(0)-ARCH(1) model .
   (a) Calculate the conditional mean, variance and kurtosis of the daily Bitcoin returns.
   (b) Plot the estimated conditional volatility of the daily Bitcoin returns. Compare the estimated conditional volatility assuming normal, student-t, skewed student-t and generalized error distributions of the standardized innovations.
   (c) Is it necessary to replace the AR(0) for the conditional mean with some alternative ARMA model?
3. Fit various types of GARCH models (GARCH(1,1), GJR-GARCH(1,1), EGARCH(1,1)) to the daily Bitcoin returns. Compare the fit of the models and the estimated conditional volatility.
4. (Optional) Fit various types of GARCH models (GARCH(1,1), GJR-GARCH(1,1), EGARCH(1,1)) to the daily S&P 500 returns. Then use the fitted model to price the SPX options in the attached file (with Monte Carlo simulation assuming appropriate distribution of the innovations) and compare with the market prices. (Hint: you can use the past year's data to fit the model and then use the fitted model to price the options.)